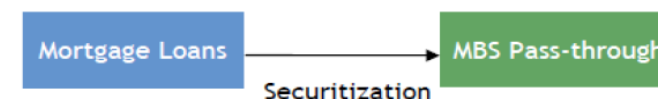
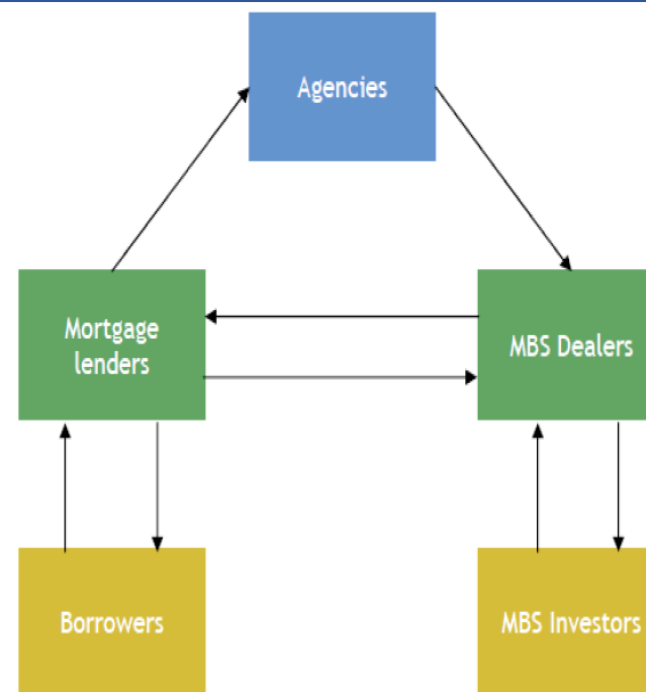
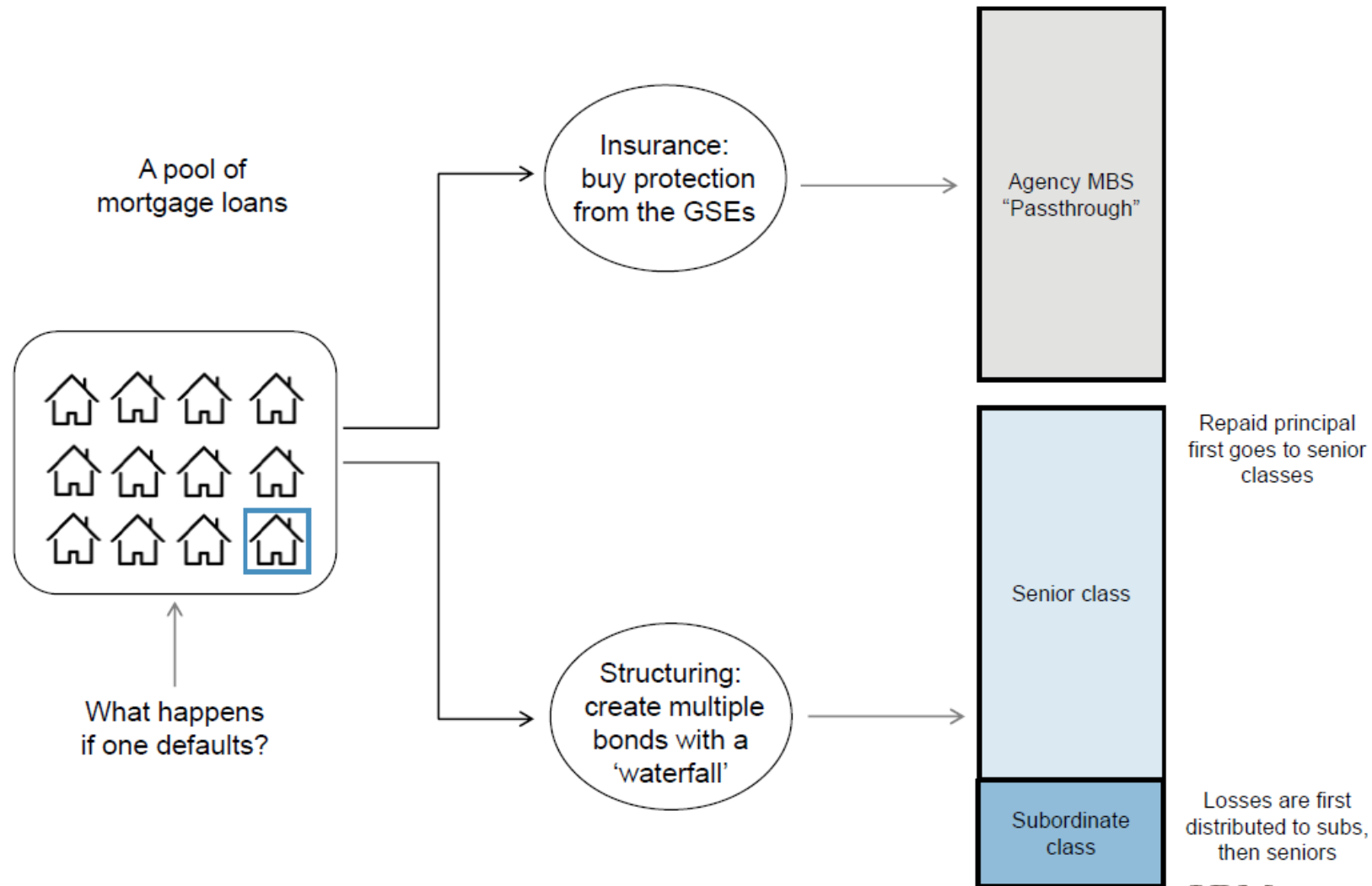

Introduction to SPG Modeling

Securitized Products Overview

- ❑ Significant portion of the Fixed Income Market
 - ❑ Several different products, e.g., MBS, ABS, CMBS, CMOs, MSR, that vary in the underlying assets, as well as complexity
 - ❑ Product usage across LOBs, e.g., CIB, CCB, CIO/Treasury, AM
 - ❑ Most import segment of the SPG market; US the most important region
 - ❑ Multitude of products, e.g., Passthrough, Spec Pools, PAC, TAC, IO, Inverse IO, Non-agency MBS, Mortgage Default Swaps/Indices
 - ❑ Varied investor base: Agencies, Banks, Pension Funds, Federal Reserve
 - ❑ Several choices available to home-owners, e.g., Fixed vs ARM vs IO, 15Y vs 30Y, Conforming vs Jumbo vs Alt-A
 - ❑ Different origination channels, e.g., Retail Banks, Brokers, Correspondents, Online platforms
-
- ❑ Modeling Considerations
 - ❑ Macroeconomic factors: Interest Rates, Housing prices, Mortgage Rates
 - ❑ Mortgages are path-dependent, and so we need a Monte-Carlo model
 - ❑ Prepayment Risk and Default risk are the primary risks associated
 - ❑ Prepayment Risk linked to rates, housing prices & other consumer choices affecting the Agency backed securities primarily

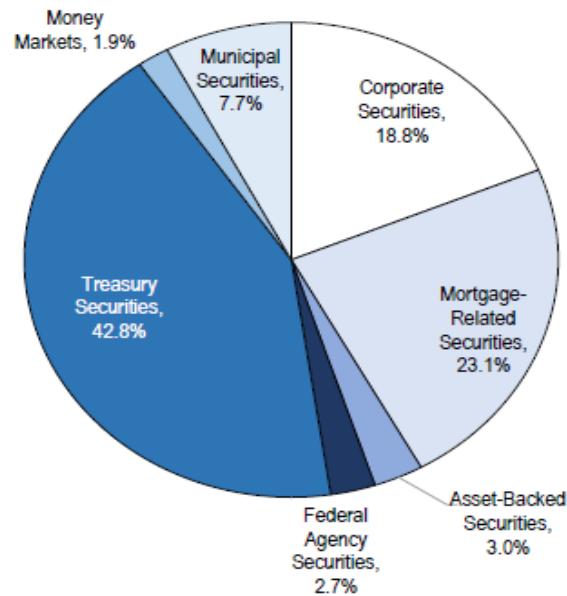


MBS Credit Protection Basics



Agency MBS In US Fixed Income Market

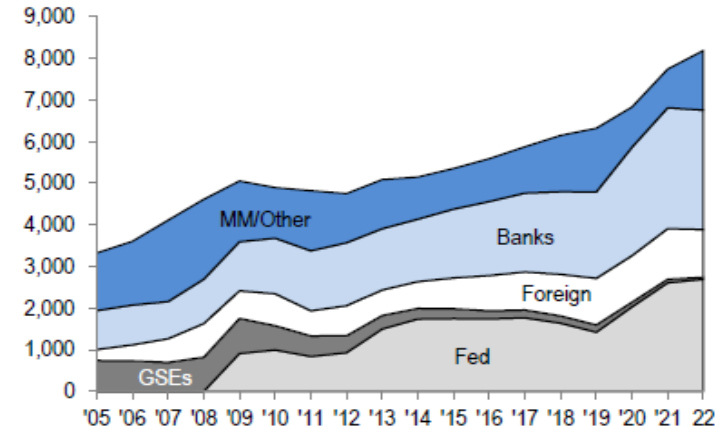
MBS in the fixed income universe (%)



Total = \$52.8 trillion

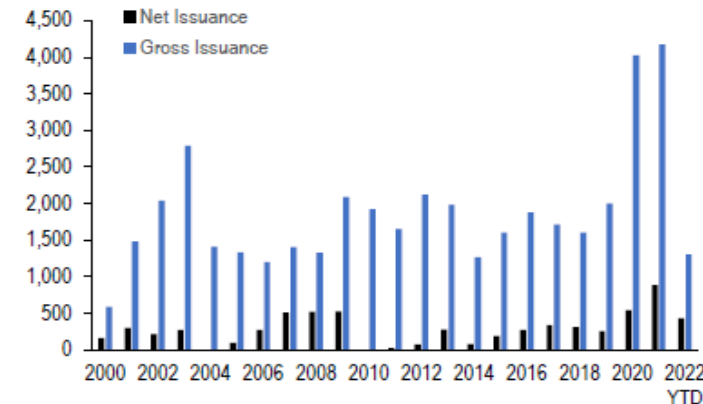
Source: Securities Industry and Financial Markets Association

Large holders of MBS (\$bn)



Source: J.P. Morgan, Federal Reserve, Ginnie Mae, Fannie Mae, Freddie Mac, company filings

Gross and net agency MBS net issuance, \$bn

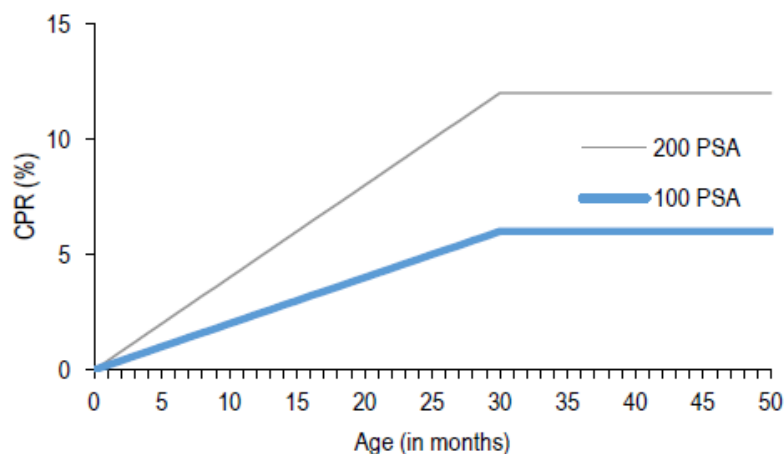


Source: J.P. Morgan

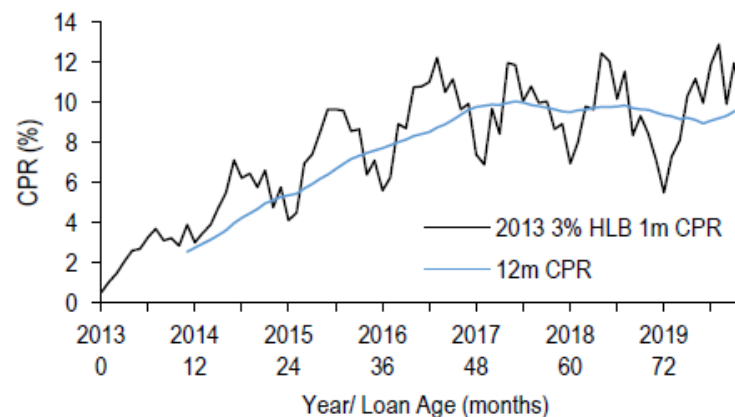
Prepayment Basics

- **SMM – Single Month Mortality** – the percentage of collateral that prepays in a single month
$$\text{SMM} = 1 - [(\text{Actual Balance}) / (\text{Scheduled Balance})]$$
$$= [(\text{Prepayment}) / (\text{Outstanding Principal})]$$
- **CPR – Conditional/Constant Prepayment Rate** – annualized percentage of remaining principal prepaid
$$\text{CPR} = 100 * [1 - (1 - \text{SMM})^{12}]$$
- **PSA – prepayment vector** expressed as a series of CPRs; begins at .2% in the first month, increases .2% per month, leveling out at 6.0% in month 30; prepayment assumptions for pricing stated as linear multiples of PSA schedule (PSA stands for Public Securities Association, a predecessor to SIFMA)

Illustrating the PSA prepayment vector



Real-life example of prepayment ramp using 2013 3% HLBs



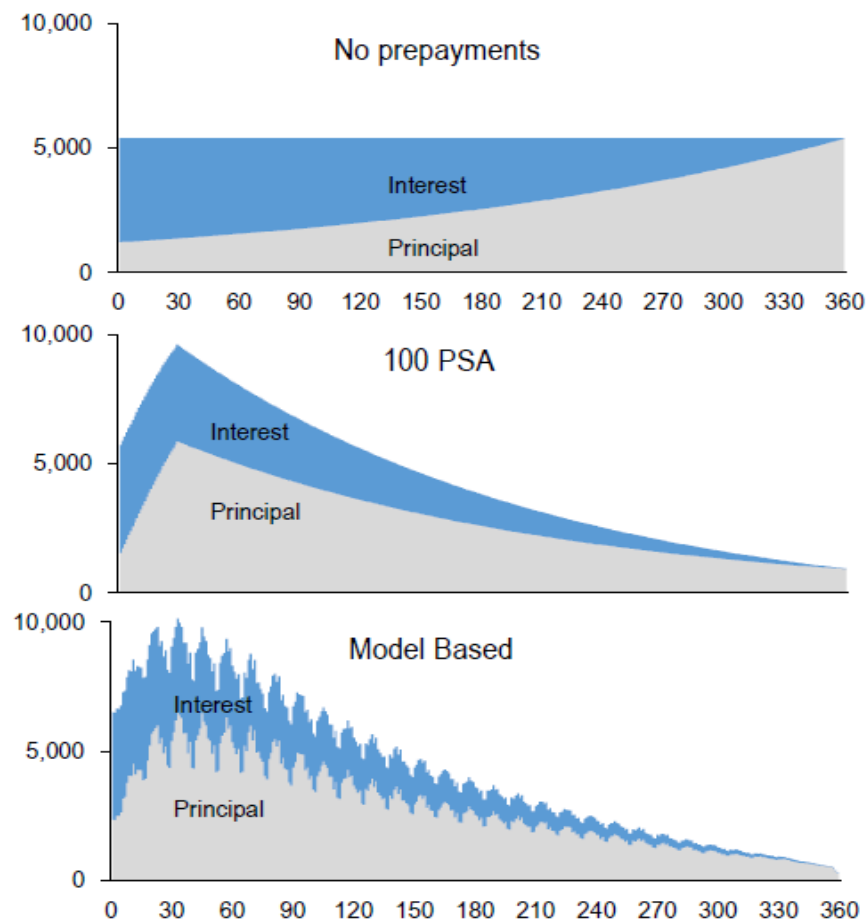
Prepayment Cashflows

MBS cashflows without prepayments are level (fully amortizing loans)

Payment Options		Payments				
Enter 2 of the next 3 fields		10 Table		11 Chart		
Loan Amount	1000000.00	Annualized Information				
Interest Rate	5.000	Payment				
Payment	5368.22		Date	Payment	Interest	Principal
Payment Freq	Monthly	1.	11/03/22	5368.22	4166.67	1201.55
		2.	12/03/22	5368.22	4161.66	1206.56
Years/Months	30 / 0	3.	01/03/23	5368.22	4156.63	1211.58
Starting	10/03/22	4.	02/03/23	5368.22	4151.58	1216.63
Points	0.000	5.	03/03/23	5368.22	4146.52	1221.70
Prepayment Options		6.	04/03/23	5368.22	4141.42	1226.79
Scheduled Prepayments		7.	05/03/23	5368.22	4136.31	1231.90
Amount		8.	06/03/23	5368.22	4131.18	1237.04
Frequency	Monthly	9.	07/03/23	5368.22	4126.03	1242.19
Starting	11/03/22	10.	08/03/23	5368.22	4120.85	1247.37
Add Additional Prepayments		11.	09/03/23	5368.22	4115.65	1252.56
		12.	10/03/23	5368.22	4110.43	1257.78
		13.	11/03/23	5368.22	4105.19	1263.02
		14.	12/03/23	5368.22	4099.93	1268.29
		15.	01/03/24	5368.22	4094.65	1273.57
		16.	02/03/24	5368.22	4089.34	1278.88
		17.	03/03/24	5368.22	4084.01	1284.21
		18.	04/03/24	5368.22	4078.66	1289.56

Source: J.P. Morgan, Bloomberg Finance L.P.

The impact of prepayments on MBS cashflows assuming \$1mm balance, 5% mortgage rate, shown as monthly payment (\$) versus loan age (months)

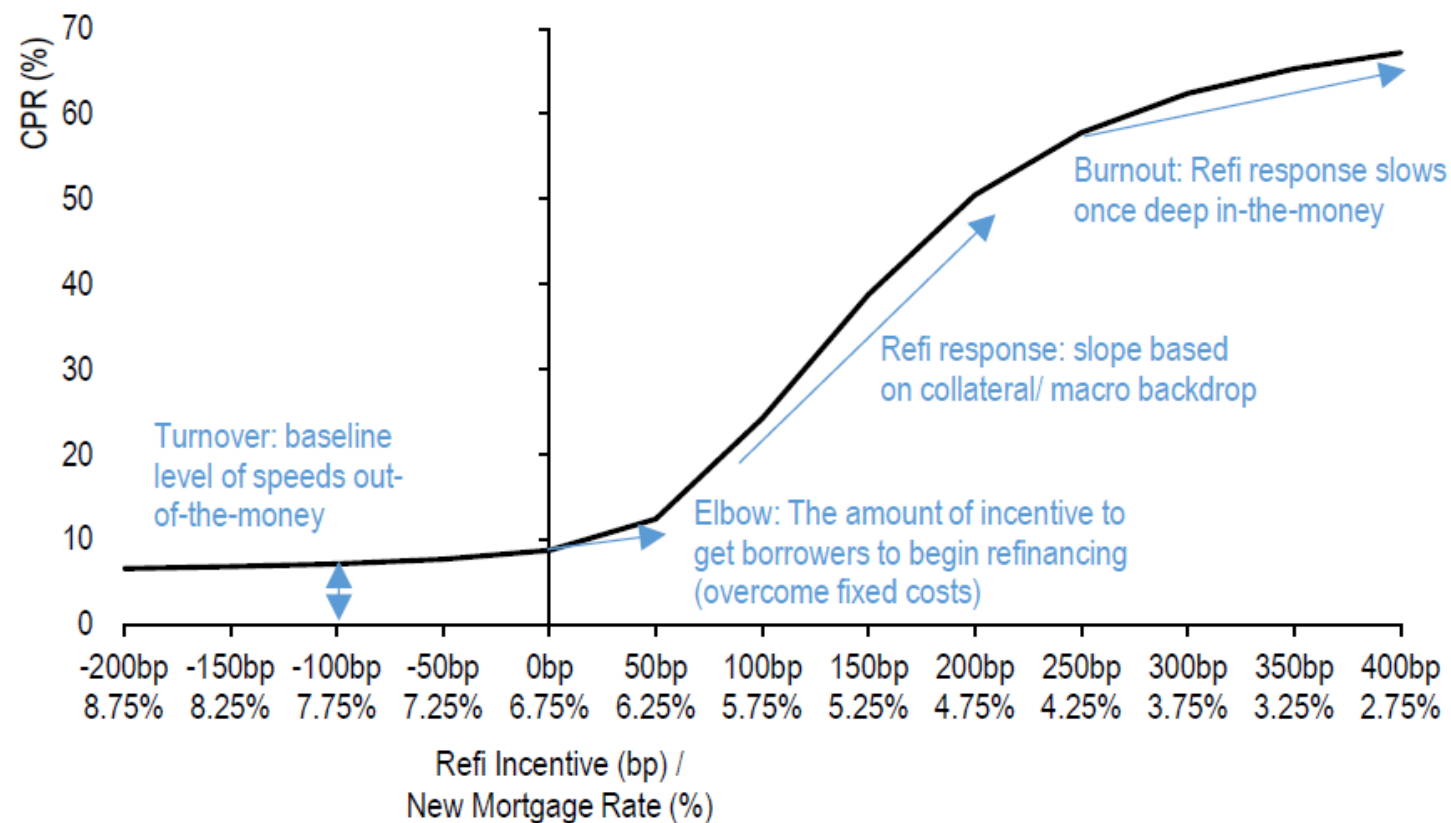


Prepayment Components

- **Prepayments are the source of MBS optionality**
 - Borrowers are long the prepayment option while MBS investors are short the option
 - ... however, unlike callable corporate bonds, borrowers do not or can not always exercise the call option efficiently
 - Prepayment analysis focuses on when and how efficiently borrowers exercise the option
- **Refinancing**
 - Largest component of prepayments
 - Borrowers take advantage of lower interest rates to refinance
 - Borrowers with accumulated equity can refinance and take out a larger mortgage
 - Cash can be used for home improvement, paying off bills, or other debt consolidation, this effect is driven primarily by home price appreciation (HPA)
- **Housing Turnover**
 - Prepayment occurs when borrower moves from one home to another
 - As loans age (or “season”) they show higher turnover speeds
 - Seasonality is an important driver of turnover, as most families move during the summer (when kids are out of school)
- **Defaults and Buyouts**
 - Borrower defaults ultimately translate into prepayments for agency investors
 - The GSEs repurchase seriously delinquent mortgages out of conventional MBS pools...
 - ... in Ginnies, servicers have the option to repurchase delinquent mortgages out of GNMA MBS (at par)
- **Curtailments**
 - Partial prepayments

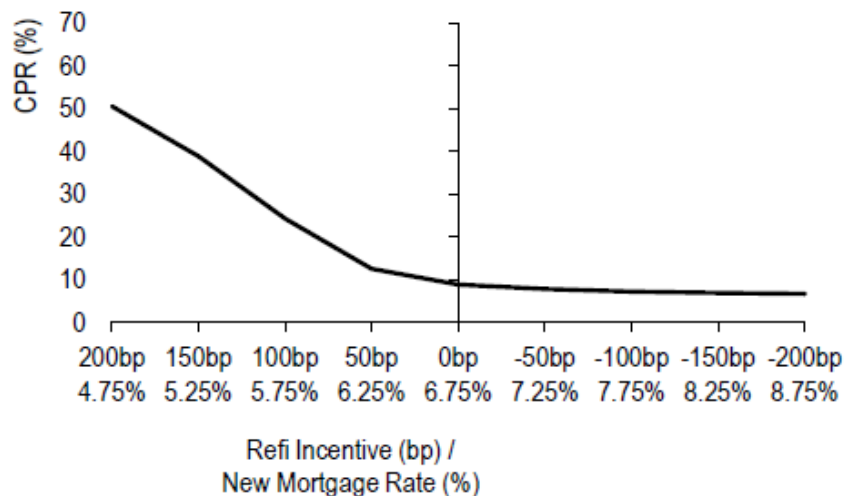
Prepayment S-Curve

Different areas of the S-curve reflect different patterns in borrower behavior,



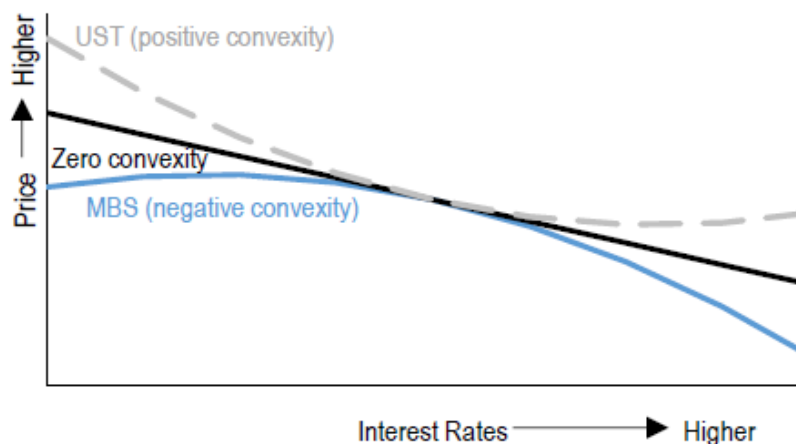
Negative Convexity

Now flip the s-curve's x-axis (interest rates) for better comparison to a standard price/rate chart...



Source: J.P. Morgan

...and consider the price performance of three fixed income products over changing interest rates

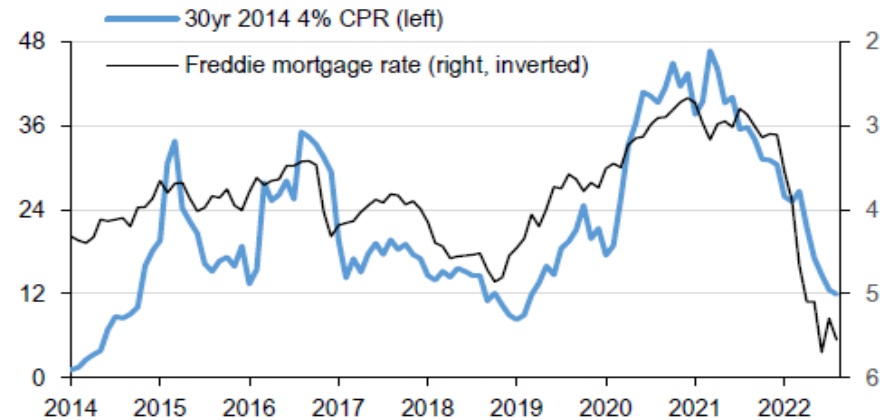


Source: J.P. Morgan

- When rates fall, prepayments pick up – so you get fewer 6% payments (assuming loan rate of 6.75%, MBS coupon of 6%) and have to reinvest at lower rates. You wanted to continue to get 6%!
- When rates go up, prepayments slow – so you get more of the 6% payments when rates are now > 6.75%. You wanted prepayments to stay high so you could reinvest at better rates!

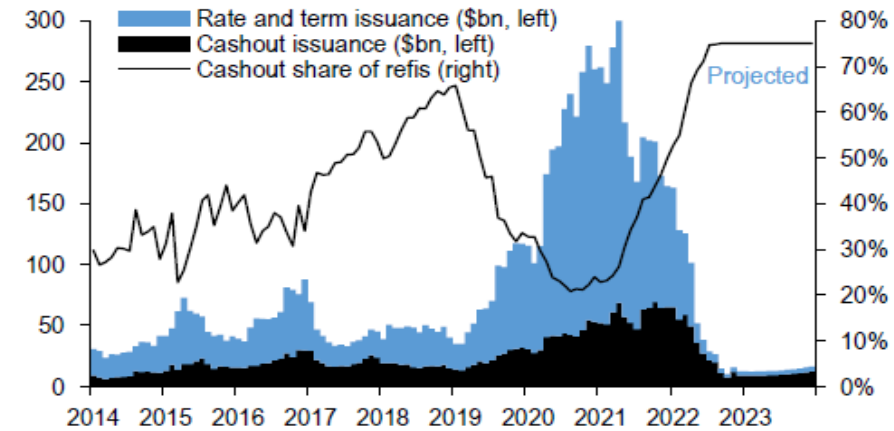
Refinancing and Cashout

Comparing the realized prepayments on 2014 4% collateral (left) against the prevailing mortgage rate (right, inverted axis)



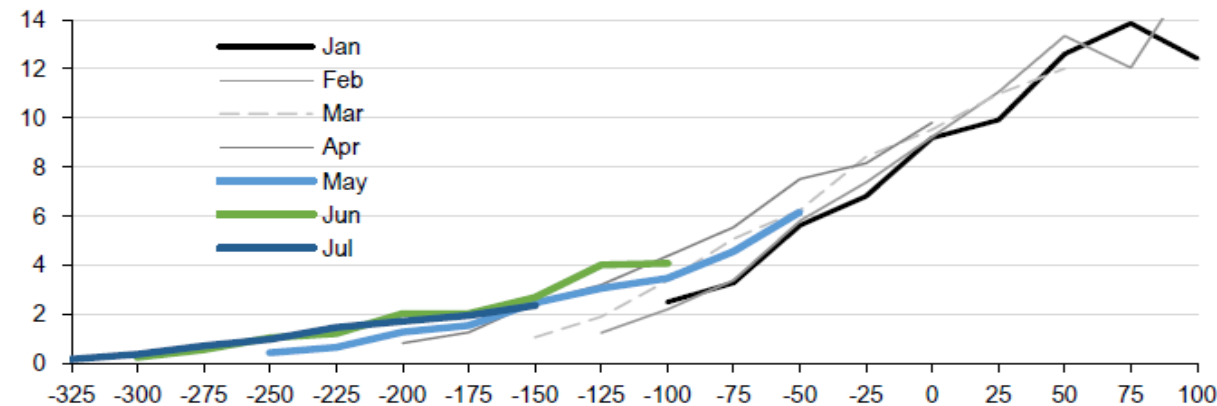
Source: J.P. Morgan, Fannie Mae, Freddie Mac

Agency issuance of cash-out versus rate and term refs (\$bn, left) and the cash-out share of the total (right)



Source: J.P. Morgan, Fannie Mae, Freddie Mac, Ginnie Mae

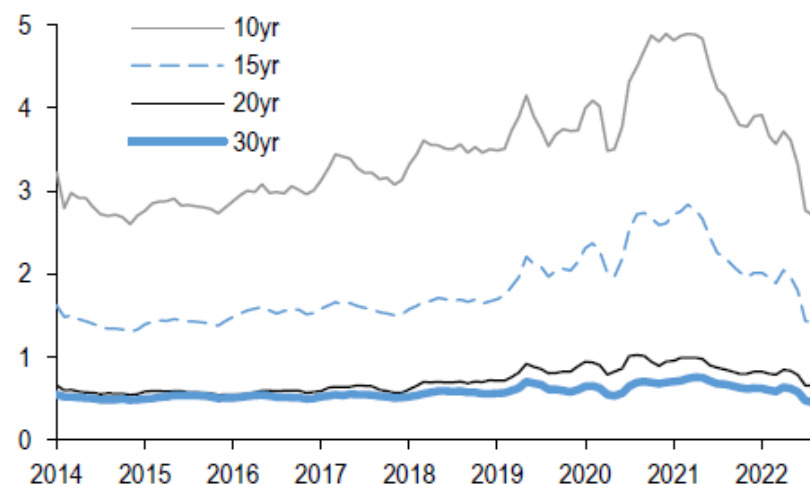
1m cash-out CPRs on conventional 30yrs, 12-36 WALA, by month in 2022, versus rate incentive (bp)



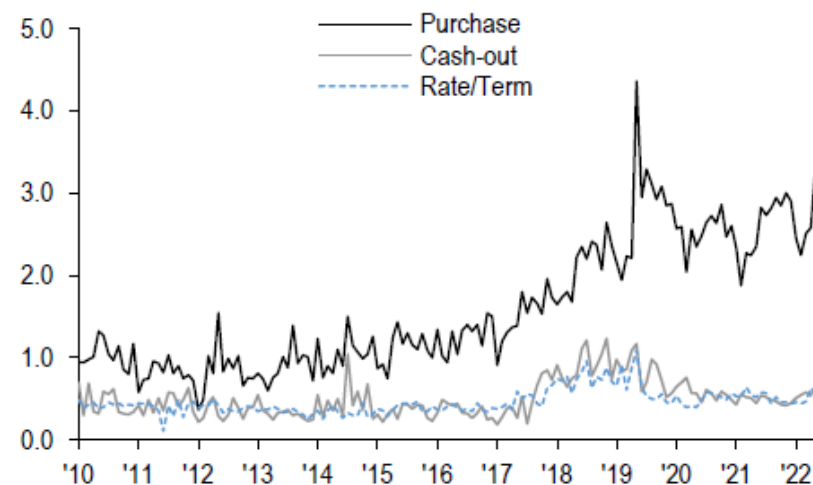
Curtailment

- Curtailments (partial prepayments) are typically the smallest portion of prepayments and are generally the most predictable given their tight correlation with product type (shorter terms curtail more), FICO (positive correlation), LTV (inverse correlation), and loan size (inverse correlation)
- Gain importance during high rate environments where small baseline speed assumptions matter for valuations
- Lately there has been a trend towards more curtailments just a few months after loan origination for purchase borrowers, most likely related to borrowers completing a sale on their previous residence and receiving an influx of cash

3m curtailment rates on 12-30 WALA conventional loans by term



FN30yr 2 WALA curtailment CPR by purpose



TBA – To Be Announced

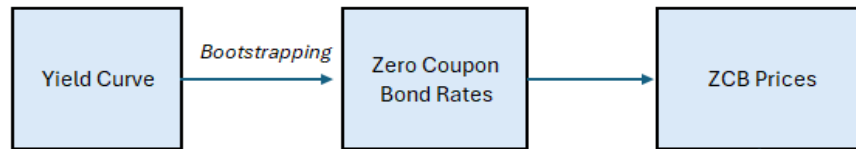
- What is a TBA?
 - A TBA is a futures contract where the type of collateral due is 'to be announced'
 - Buyer agrees to take delivery of MBS with a particular **Term, Issuer, Coupon – on a particular day (1 per Delivery Month)**
 - (for example, 30-year Fannie Mae 3s for October settle)
 - Term: 30yr/15r, Issuer: UMBS/Ginnie, Coupon: 1.5-6.5s (varies), Month: typically 3 settles at any given time
 - Seller can decide which of many existing pools of collateral to deliver (WAC, vintage, loan size, etc.) into the TBA contract
 - Allows very large trades & huge volume (\$200bn/day)
 - TBA trades settle on 1 day per month (a.k.a. PSA settle)

Characteristics of recently-issued 30-year Fannie Mae 3% MBS pools that are deliverable into a Fannie Mae 3% TBA

Issue Date	Pool Number	Cbal, \$bn	WAC	WALA	Loan size, \$k	FICO	% LTV	1m CPR	OAS
Jul 22	FN MA4698	0.5	3.62	3	399	761	78	2	31
Jun 22	FN MA4653	1.8	3.63	4	403	758	73	2	32
May 22	FN MA4624	3.3	3.65	4	389	759	73	3	33
Apr 22	FN MA4599	8.5	3.81	5	390	756	76	2	33
Mar 22	FN MA4579	13.5	3.84	6	402	748	76	4	36
Feb 22	FN MA4564	5.0	3.75	7	391	722	78	5	38
Jan 22	FN MA4549	1.9	3.73	8	388	713	79	4	39
Dec 21	FN MA4513	0.9	3.84	9	322	698	78	14	44
Nov 21	FN MA4494	0.5	3.82	10	333	698	79	10	45
Jun 21	FN MA4380	0.8	3.86	15	312	710	78	13	47
May 21	FN MA4357	0.8	3.85	16	324	710	79	9	48
Aug 20	FN MA4121	0.9	3.86	25	317	733	76	13	52
Jul 20	FN MA4097	1.6	3.86	26	316	733	77	13	53
Jun 20	FN MA4079	1.5	3.85	27	321	738	76	12	53
May 20	FN MA4048	1.6	3.85	28	304	741	77	11	54
Apr 20	FN MA4020	4.1	3.88	29	291	741	77	11	54
Mar 20	FN MA3991	2.9	3.87	30	317	751	75	13	53

The TBA typically prices to the 'worst-to-deliver' or 'cheapest-to-deliver'

Rate Simulation



$$r_{t+\Delta t} = r_t + \alpha(\gamma - r_t)dt + \sigma \varepsilon_{t+\Delta t} \sqrt{\Delta t}$$

where,

$$\varepsilon_{t+\Delta t} \sim N(0,1), t = 0, \dots, T - \Delta t$$

ZCB rate to ZCB price relation

$$r(T) = \frac{\ln\left(\frac{1}{P(T)}\right)}{T}$$

derived from,

$$P(T) = e^{-r(T)T}$$

Evolution of ZCB prices at future time t (Closed Form)

$$P(t, T) = A(t, T) e^{-B(t, T)r(t)}$$

where,

$$A(t, T) = \exp \left[(B(t, T) - T + t) \left(\frac{\alpha^2 \gamma - \frac{\sigma^2}{2}}{\alpha^2} \right) - \frac{\sigma^2 B(t, T)^2}{4\alpha} \right]$$

and,

$$B(t, T) = \frac{1 - e^{-\alpha(T-t)}}{\alpha}$$

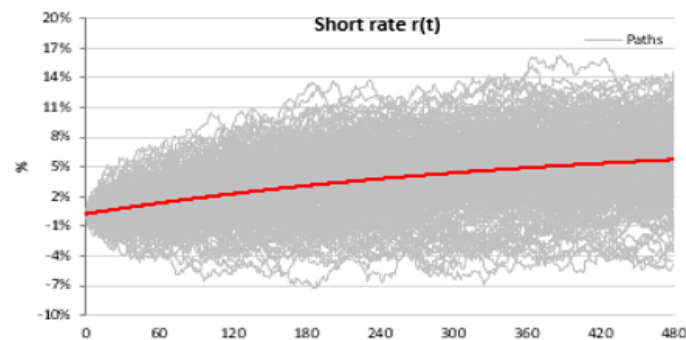
Fit parameters
Vasicek / Hull-White Model

Simulate
Instantaneous rates
 $r(t)$

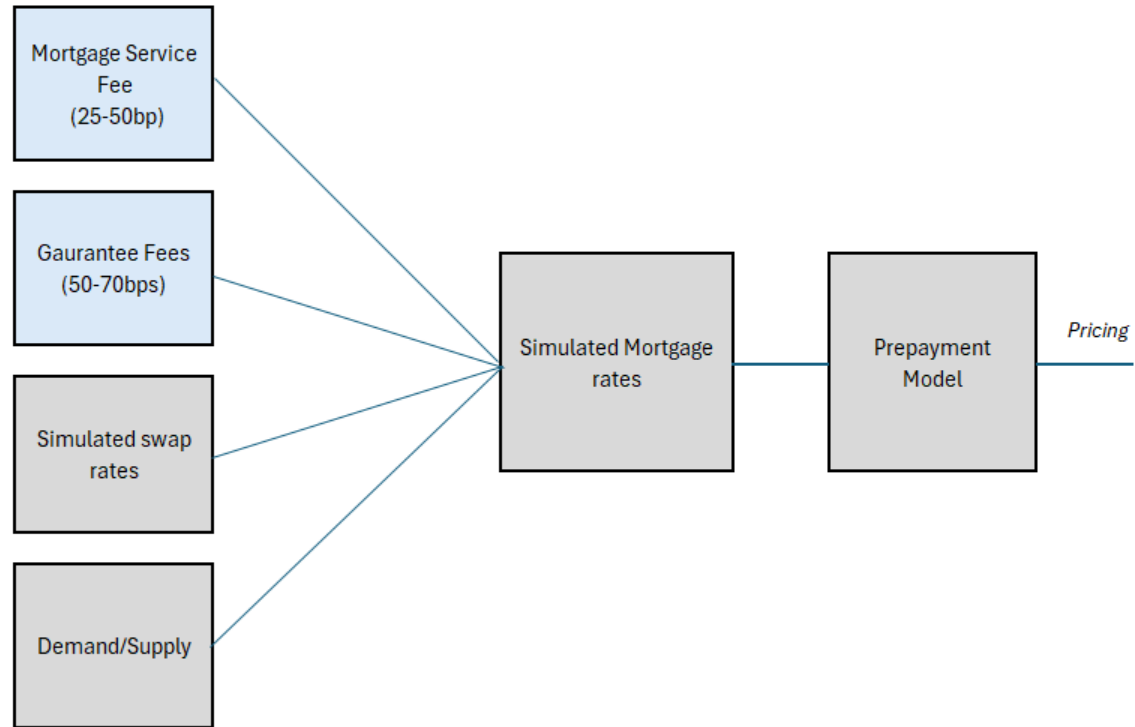
Generate Swap rates

Free parameters	
α	Mean-reversion
γ	Long term rate
Market Implied	
σ	Market Implied Vol
r_0	short rate at $t = 0$

$$S_{\alpha, \beta} = \frac{P(t, T_\alpha) - P(t, T_\beta)}{\sum_{i=\alpha+1}^{\beta} \tau_i P(t, T_i)}$$



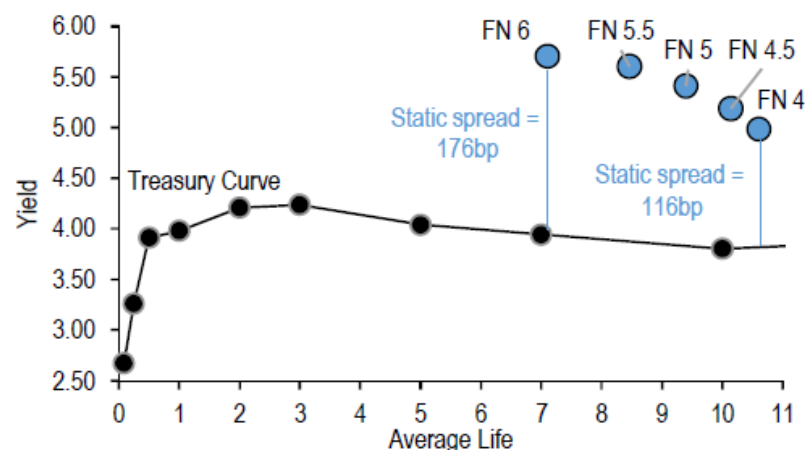
Mortgage Rates



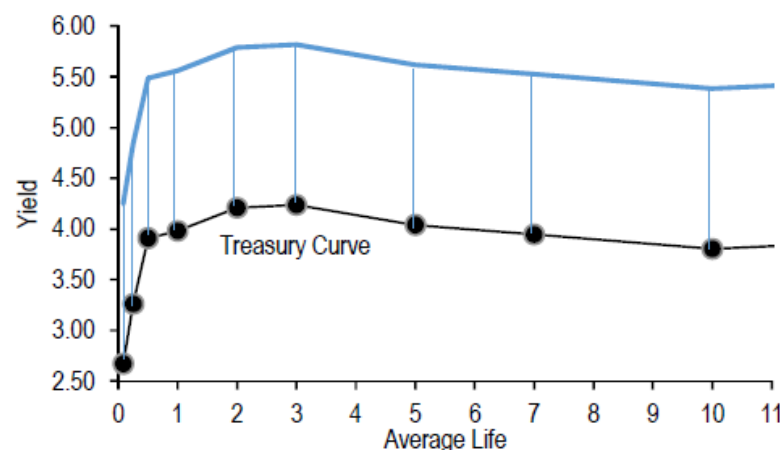
MBS Spreads

- **Static Spread (Yield Spread):** standard measure of incremental return over a single benchmark Treasury
 - Compares MBS to single point on the yield curve, usually to the interpolated point closest to the Weighted Average Life of the MBS
 - But MBS does not return principal in one lump sum but over many periods. A better assumption would include multiple data points on the yield curve. ZV Spread takes this another step further
- **ZV Spread (Yield Curve Spread):** discounts each monthly MBS cashflow by the monthly forward rates derived from the current yield curve
 - More accurate for securities that return principal over many periods as opposed to bullets
 - Still a static measure since it assumes that interest rates and MBS cashflows remain constant

Static spreads are to a single point on the nominal Treasury yield curve



A spread to the entire curve better captures monthly mortgage cashflows



Note: this chart is for illustrative purposes only. ZV spreads are actually computed to the monthly discount rates along the forward curve derived from these Treasury benchmarks, not to the benchmark yields themselves.

OAS – Option Adjusted Spread

- **OAS employs a Monte-Carlo interest rate simulation to estimate the value of the option**

- By convention, $ZV - \text{Option Cost} = \text{OAS}$
- Hundreds of hypothetical interest rate paths are simulated – need a model to simulate rate paths
- On each interest rate path the prepay model is used to predict prepayment speeds and MBS cashflows
- OAS is the single spread to forward rates across all paths that would produce prices that average to today's

- **Drawback of OAS**

- The spread earned by the investor depends on the actual path realized and can be drastically different from the OAS
- Wide differences in OASs are produced by different firms' models due to different term structures, volatility assumptions and prepayment projections
- Does not account for dollar roll financing
- Is a “black box” – difficult for investors to decompose OAS into its component parts

Greeks

- **Every model produces its own estimates of MBS greeks**
 - Models even have slightly different ways of estimating greeks within the OAS framework
 - The general concept is to hold OAS constant while applying shocks to rates, etc.
- **OAD / OAC**
 - Hold OAS constant, shock rates +/- 10bp
- **Vega**
 - Hold OAS constant, shock vol surface up/down 1abp (calibrated 10yr tails, in our model)
- **Spread duration**
 - Shock OAS up/down 10bp, see price impact

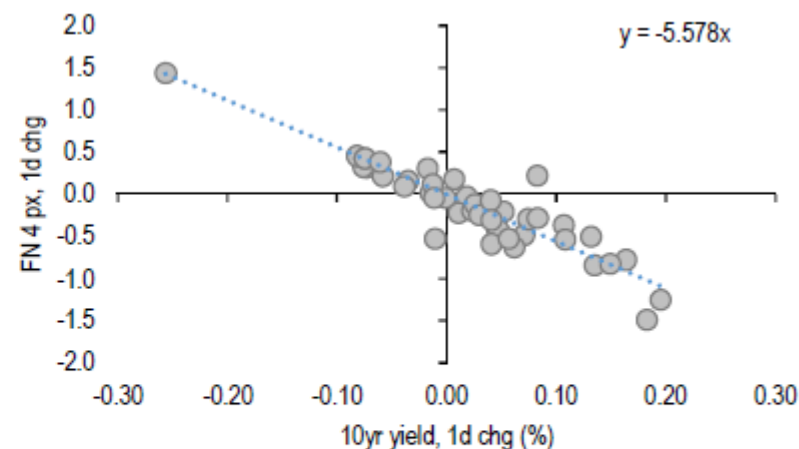
Duration

■ Empirical durations

- Simple to construct, transparent: rolling regressions vs. rates
- TBA assumption & model independent
- Backward looking, struggle in sharp rate moves
- Can erroneously pick up spread moves as duration

■ Model durations (Option Adjusted Durations)

- Forward looking, takes collateral into account, allowing for dynamic projections in different scenarios
- Can better account for shifts in the TBA deliverable
- Unfortunately, models do not always capture how the market is trading
- Allows for estimation of **partials** or **key rates**

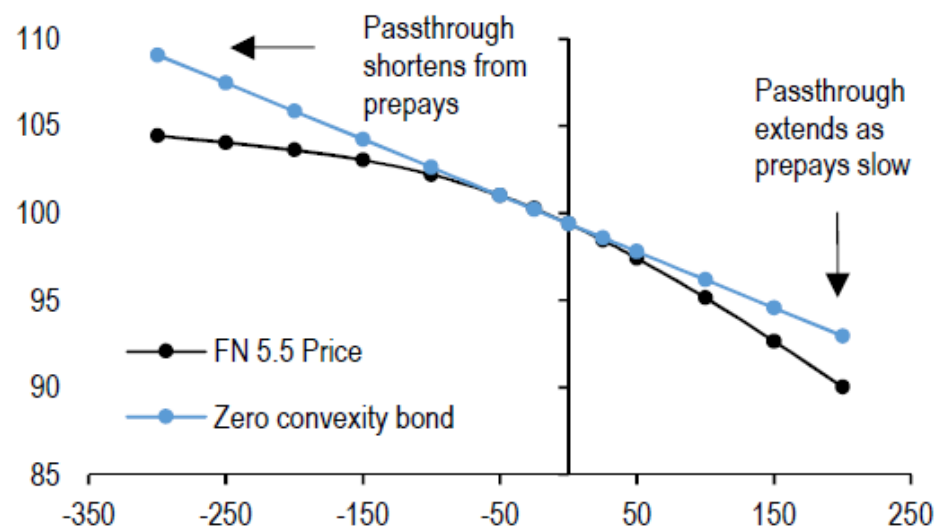


Source: J.P. Morgan

Convexity

- Convexity: the rate at which the duration of a security changes as interest rates change
 - Positive convexity implies that for large, equal and opposite changes in interest rates, the increase in price if rates go down will be more than the decrease in price if rates rise
 - Negative convexity implies that the increase in price if rates go down will be smaller than the decrease in price if rates rise
 - Bullet Treasuries have positive convexity. Pass-throughs typically have negative convexity

FN 5.5 prices (\$) vs. shift in rates (bps)



Source: J.P. Morgan

Duration and convexity across the stack (bps)

		Duration	Convexity
UMBS 30	1.5	7.6	0.5
	2	7.6	0.3
	2.5	7.2	0.1
	3	7.0	0.0
	3.5	6.5	-0.3
	4	5.9	-0.9
	4.5	5.1	-1.1
	5	4.3	-1.4
	5.5	3.6	-1.5
	6	2.8	-1.5

Source: J.P. Morgan

THANK YOU